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Case 12-2026: An 86-Year-Old Woman with Anorexia, Weight Loss, and Liver Lesions

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PRESENTATION OF CASE

Dr. Dustin G. Shaw (Medicine): An 86-year-old woman presented to the emergency department of this hospital because of anorexia, weight loss, and weakness.

The patient had been in her usual state of health until 3 weeks before the current presentation, when anorexia developed with minimal oral intake. The patient noticed generalized weakness, and she had difficulty getting out of bed and performing activities of daily living. Owing to progressively worsening symptoms, she was brought by a friend to the emergency department of this hospital.

On examination in the emergency department, the heart rate was 85 beats per minute, the oral temperature 35.9°C, the blood pressure 113/56 mm Hg, the respiratory rate 16 breaths per minute, and the oxygen saturation 97% while the patient was breathing ambient air. Physical examination revealed mild conjunctival icterus, fine basilar crackles in the lungs, and tenderness on deep palpation of the right upper quadrant without rebound or guarding. No organomegaly or adenopathy was detected.

Dr. Rory L. Cochran: Computed tomography (CT) of the abdomen and pelvis, performed after the administration of intravenous contrast material (Fig. 1A and 1B), revealed choledocholithiasis, thickening of the bile duct walls, and mild intrahepatic and extrahepatic ductal dilatation. In addition, four indeterminate hypodense hepatic lesions were identified in the left and right lobes of the liver. One lesion in liver segment 4 measured 4.5 cm in the greatest dimension.

Dr. Shaw: Intravenous lactated Ringer's solution and oral potassium chloride were administered. The patient was subsequently admitted to the hospital.

A review of systems was notable for anorexia and a weight loss of 7 kg during the previous 3 weeks. The patient reported no fever, rigors, night sweats, abdominal pain, nausea, pruritus, darkening of the urine, hematochezia, melena, acholic stools, or postprandial abdominal discomfort. Her medical history was notable for hyperlipidemia, type 2 diabetes mellitus, long-standing nonprogressive fibrotic lung

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CME



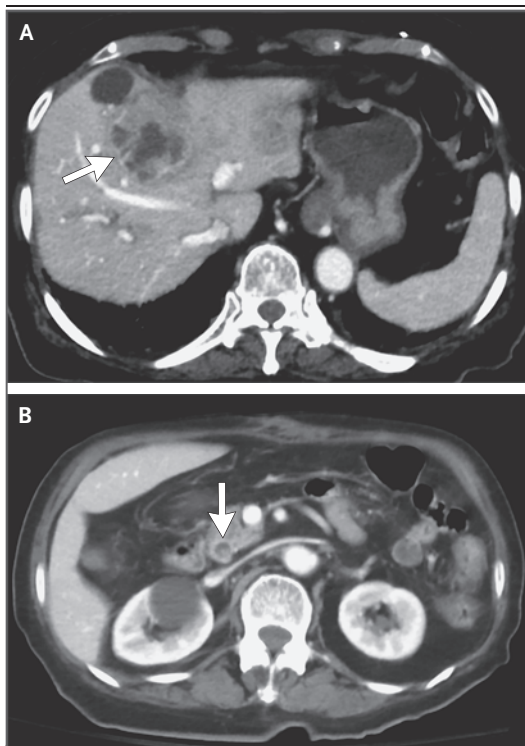


Figure 1. CT of the Abdomen and Pelvis.

An axial image of the abdomen and pelvis, obtained after the administration of intravenous contrast material (Panel A), shows a heterogeneous, hypodense intrahepatic lesion in segment 8 (arrow). An axial contrast-enhanced image obtained at the level of the distal common bile duct (Panel B) shows ductal dilatation and choledocholithiasis (arrow).

disease, mild cognitive impairment, and localized estrogen receptor–positive ductal carcinoma that had been treated with mastectomy and tamoxifen therapy 30 years earlier. Medications included metformin, donepezil, and rosuvastatin. She had no known adverse reactions to medications. She was originally from East Asia but had lived in urban Massachusetts for decades, with no recent international travel. She was widowed, lived with a friend, and had a pet cat. Her family history was notable for gallstones in her mother.

Laboratory test results were notable for a potassium level of 3.0 mmol per liter (reference range, 3.4 to 5.0), an alkaline phosphatase level of 418 U per liter (reference range, 45 to 115), and a total bilirubin level of 2.2 mg per deciliter (38 μ mol per liter; reference range, 0.0 to 1.0 mg per deciliter [0 to 17 μ mol per liter]). The blood levels of carcinoembryonic antigen, CA 19-9, and CA 15-3 were within normal limits. Blood specimens were obtained for culture, and no growth was observed; additional laboratory test results are shown in Table 1.

Dr. Cochran: Magnetic resonance imaging (MRI) of the abdomen and magnetic resonance cholangiopancreatography (MRCP) were performed on hospital day 1 (Fig. 2A, 2B, and 2C). MRCP confirmed the presence of choledocholithiasis and showed biliary ductal dilatation with wall thickening and hyperenhancement. The four hepatic

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Initial Evaluation
Blood		
Hemoglobin (g/dl)	12.0–16.0	13.1
Hematocrit (%)	36.0–46.0	41.3
White-cell count (per μ l)	4500–11,000	12,260
Differential count (per μ l)		
Neutrophils	1800–7700	10,370
Lymphocytes	1000–4800	1080
Monocytes	200–1200	680
Eosinophils	0–900	10
Basophils	0–300	10
Immature granulocytes	0–100	20

Table 1. (Continued.)

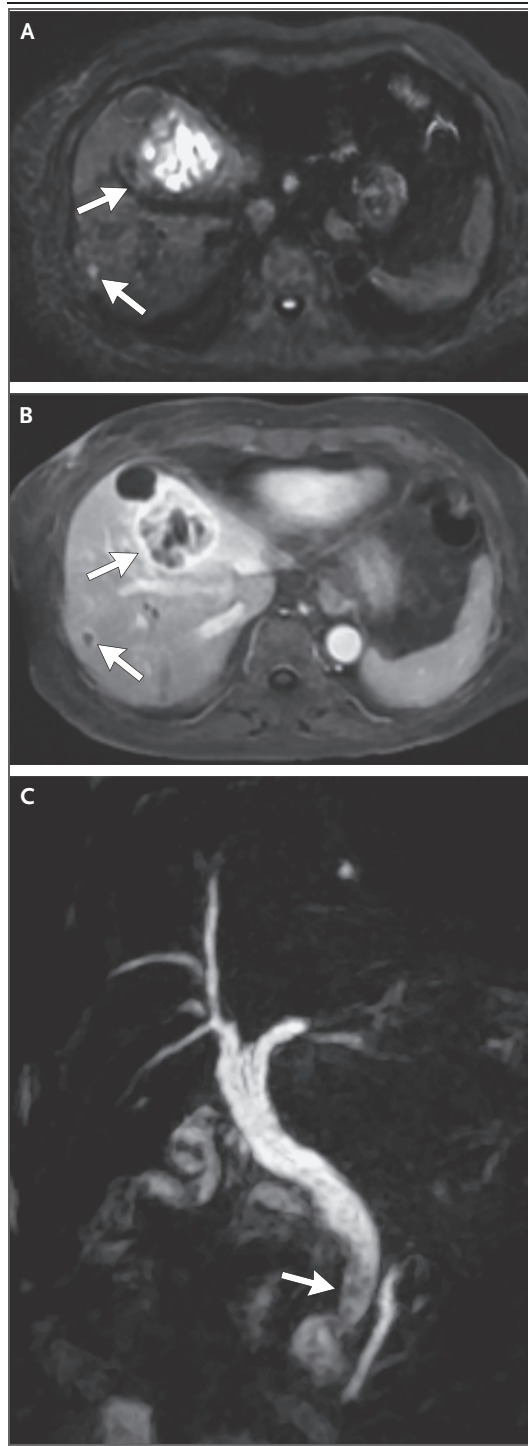
Variable	Reference Range, Adults [†]	On Initial Evaluation
Platelet count (per μ l)	150,000–400,000	311
Sodium (mmol/liter)	135–145	132
Potassium (mmol/liter)	3.4–5.0	3.0
Chloride (mmol/liter)	98–108	93
Carbon dioxide (mmol/liter)	23–32	24
Urea nitrogen (mg/dl)	8–25	20
Creatinine (mg/dl)	0.60–1.50	0.60
Glucose (mg/dl)	70–110	165
Phosphorus (mg/dl)	2.6–4.5	2.8
Calcium (mg/dl)	8.5–10.5	10.4
Total protein (g/dl)	6.0–8.3	7.4
Albumin (g/dl)	3.3–5.0	3.4
Alanine aminotransferase (U/liter)	7–33	28
Aspartate aminotransferase (U/liter)	9–32	32
Alkaline phosphatase (U/liter)	45–115	418
Total bilirubin (mg/dl)	0.0–1.0	2.2
Direct bilirubin (mg/dl)	0.0–0.4	1.5
International normalized ratio	0.91–1.16	1.10
Urine		
Color	Yellow	Yellow
Clarity	Clear	Clear
pH	5.0–9.0	6.5
Specific gravity	1.001–1.035	1.040
Glucose	Negative	1+
Ketones	Negative	3+
Leukocyte esterase	Negative	Negative
Nitrite	Negative	Negative
Blood	Negative	Negative
Protein	Negative	1+
Erythrocytes (per high-power field)	0–2	0
Leukocytes (per high-power field)	<10	0
Bacteria	None	None

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for phosphorus to millimoles per liter, multiply by 0.3229. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

[†] Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

lesions previously identified in segments 2, 4, 7, and 8 showed heterogeneous signal intensity on T2-weighted images, irregular enhancement, and restricted diffusion.

Dr. Shaw: A diagnostic test was performed.



DIFFERENTIAL DIAGNOSIS

Dr. Margaret M. Chapman: This 86-year-old woman presents with anorexia, weight loss, and liver lesions. In constructing a differential diagnosis for this case, I will use several core principles of clinical reasoning: illness scripts, problem representation, and dual process theory.¹ As physicians, we must be cognizant of our initial tendencies when approaching a case, since early, fast thinking places us at risk for reaching quick conclusions, potentially leading to an incorrect diagnosis. By taking a methodical approach and following the structure of a problem representation statement, I will explore some of the nuances of this patient's case to reach a presumptive diagnosis.

One way to approach this patient's case would be to focus on the multiple liver lesions, which are the most unique and differentiating feature of her presentation. This strategy will initially expedite the diagnostic process to ensure a broad differential diagnosis before incorporating the patient's additional symptoms. The differential diagnosis for liver lesions, irrespective of a patient's epidemiologic risk factors, includes benign nodular and cystic liver lesions, cancer, infection, and vascular malformations.

On the basis of the features of this patient's presentation, I would frame the case as that of an 86-year-old woman with type 2 diabetes mellitus and remote breast cancer who presents with a subacute illness characterized by wasting, cholestatic liver injury, and liver lesions. I will use the components of this problem representation to narrow the differential diagnosis.

Figure 2. MRI of the Abdomen and MRCP Image.

MRI of the abdomen and magnetic resonance cholangiopancreatography (MRCP) were performed on hospital day 1. An axial intermediate b-value diffusion-weighted image (Panel A) and a T1-weighted, fat-saturated image obtained after the administration of intravenous contrast material (Panel B) show hepatic lesions in segments 4 and 7 with restricted diffusion and irregular enhancement, respectively (arrows). A coronal oblique maximum-intensity-projection reformatted image obtained from MRCP (Panel C) shows choledocholithiasis in the distal common bile duct (arrow) and intrahepatic and extrahepatic biliary ductal dilatation.

DEFINING THE SUBSTRATE

It is critical to consider relevant patient-specific risk factors for diseases, and in clinical reasoning, this is often referred to as defining the substrate. In this patient, I would highlight her advanced age, breast cancer history, and type 2 diabetes mellitus. She also has fibrotic lung disease, but this condition is unlikely to be contributing to her current presentation. Focusing on these details allows us to further narrow the differential diagnosis by ruling out irrelevant diagnoses.

NODULAR REGENERATIVE HYPERPLASIA AND POLYCYSTIC LIVER DISEASE

Conditions associated with nodular liver lesions, such as nodular regenerative hyperplasia, are less common in older adults than in younger persons, and regenerative nodules often develop in the context of cirrhosis,² which this patient does not have. Therefore, we can rule out conditions associated with nodular liver lesions.

Polycystic liver disease is asymptomatic in 95 to 98% of affected persons, but when cystic lesions cause symptoms, they typically result from capsular stretch or compression of adjacent structures.³ Although polycystic liver disease can occur in women of advanced age, the characteristic cystic lesions are not consistent with the imaging findings in this patient. The presence of simple hepatic cysts is similarly unlikely in this patient on the basis of her imaging results, despite the prevalence of these cysts among older adults.⁴

CANCER

At first glance, one could quickly assume that cancer is responsible for the patient's weight loss, malaise, and multifocal liver lesions. Both her remote history of breast cancer and her advanced age increase her risk of new or recurrent cancer. Breast cancer metastasizes to the liver in approximately 50% of patients with metastatic disease. Although late recurrence with metastatic disease can occur, it would be very unusual for a recurrence to arise 30 years after the initial diagnosis and treatment.⁵ New cancer could be present. Possible considerations include primary hepatobiliary neoplasm (e.g., hepatocellular carcinoma or cholangiocarcinoma); a metastatic lesion from a distant site such as the gastrointestinal tract, lung, or pancreas; and melanoma.⁶ However, the short 3-week duration of the patient's illness is incom-

patible with the tempo of most cancers that cause liver lesions, since it takes time for the necessary cellular replication to form a malignant lesion large enough to cause the degree of symptoms that is seen in this patient. Thus, the pace of her illness points us away from a diagnosis of cancer and toward a diagnosis of infection or vascular malformations.

NAMING THE CLINICAL SYNDROME

The final component of the problem representation is to name the clinical syndrome. In addition to the multifocal liver lesions, the patient's clinical course is also characterized by wasting, with a 7-kg weight loss, and cholestatic liver injury, each of which suggests an inflammatory process leading to liver injury and tissue catabolism. When considering these features and the liver lesions on cross-sectional imaging, I can quickly rule out other diagnoses, including vascular malformations and liver tumors. Vascular malformations and tumors of the liver are often clinically silent and would be unlikely to produce the degree of wasting that is seen in this patient.^{7,8}

NONPYOGENIC AND ECHINOCOCCAL LIVER ABSCESES

Ultimately, infection is the most likely diagnosis in this patient's case. Liver lesions due to infection can be broadly grouped into pyogenic abscesses, nonpyogenic abscesses, and echinococcal cysts. On the basis of this patient's epidemiologic risk and imaging features, we can rule out nonpyogenic liver abscess and echinococcal cysts. Nonpyogenic liver abscess may be caused by infection with *Entamoeba histolytica*, which typically occurs in young men. A liver abscess caused by *E. histolytica* infection most often manifests as a solitary lesion. In the United States, approximately 20% of liver abscesses are caused by this pathogen.⁹ This patient has lived in urban Massachusetts without any recent epidemiologically significant travel, but she could have had possible exposures during her time in East Asia.

Although the patient's exposure risk in East Asia is sufficiently remote to reduce concern about the development of amebiasis now, echinococcal cysts can persist for decades before causing symptoms, which usually occur because of mass effect or spontaneous rupture with subsequent anaphylaxis.¹⁰ However, echinococcus species are not as

common in East Asia as in other regions. Furthermore, this patient's lesions are not cystic, she does not have peripheral eosinophilia, and she is quite symptomatic. All these features make the diagnosis of liver disease due to echinococcal infection unlikely. As such, I will center my approach on pyogenic multifocal liver abscess, since this diagnosis aligns with the patient's history and examination and cross-sectional imaging findings.

PYOGENIC LIVER ABSCESS

Multiple mechanisms can lead to the development of pyogenic abscess. Of these, the most relevant to this patient's case include ascending biliary infection, venous drainage from an intraabdominal infection, hematogenous spread from a bloodstream infection, and direct extension from an adjacent site of infection. To compare these four scenarios, I will focus on distinguishing features of each. Biliary infection accounts for the majority of cases of pyogenic abscess.⁹ In these cases, gallstones or other causes of obstruction of the common bile duct, with or without overt cholangitis, can serve as an anchor point for ascending polymicrobial infection and subsequent abscess formation.

Pyogenic abscesses that form from a nonbiliary source can develop as a result of direct extension of an adjacent infection, such as cholecystitis, a perinephric abscess, or a subphrenic abscess. Monomicrobial abscesses stemming from hematogenous spread typically occur in patients with bacteremia. Enteric organisms, *Staphylococcus aureus*, and streptococcus species are the most common causative bacteria. More recently, infection with hypervirulent strains of *Klebsiella pneumoniae* has become an important cause of liver abscesses resulting from hematogenous spread, especially in patients from East and Southeast Asia.¹¹ Finally, a distant intraabdominal infection, such as smoldering diverticulitis or colitis, can cause pyogenic abscess through portal venous drainage and typically affects the right lobe of the liver.

By comparing the illness scripts of these four causes of pyogenic abscess with this patient's history, negative blood cultures, and findings on cross-sectional imaging, I suspect that an ascending biliary infection is the most likely diagnosis, given evidence of cholestasis and gallstones on imaging. To confirm the diagnosis of pyogenic

liver abscess and to rule out the presence of liver metastases or a primary hepatobiliary tumor, I would recommend performing a liver biopsy for pathological evaluation and culture.

DR. MARGARET M. CHAPMAN'S DIAGNOSIS

Pyogenic liver abscess infection, most likely originating from the biliary tract.

DIAGNOSTIC STUDIES AND DISCUSSION OF MANAGEMENT

Dr. Kumar Krishnan: Given the detection of choledocholithiasis on MRCP, endoscopic retrograde cholangiopancreatography (ERCP) was planned. In addition, the presence of multiple liver masses on imaging was suggestive of metastases and warranted further evaluation with endoscopic ultrasonography and guided fine-needle biopsy for pathological confirmation.

An initial upper endoscopy was performed (Fig. 3A). After no abnormal findings were noted in the esophagus, stomach, or duodenum, the therapeutic linear echoendoscope was advanced to the stomach. From the gastric cardia, a round mass was identified in the left lobe of the liver. The mass was lobulated, appeared multicystic, and measured 30 mm in the maximal cross-sectional diameter. Fine-needle biopsy of the mass was performed with the use of a 22-gauge core-biopsy needle.

Next, ERCP was performed (Fig. 3B). After cannulation of the bile duct, endoscopic ultrasound images showed a heterogeneous, well-circumscribed mass in the left lobe of the liver (Fig. 3C) and a moderately dilated common bile duct with numerous filling defects that were consistent with gallstones (Fig. 3D). A complete biliary sphincterotomy was performed. Extensive sludge and stone fragments were removed with the use of an extraction balloon.

PATHOLOGICAL DISCUSSION

Dr. Ahmad Alkashash: A core-biopsy specimen of the liver mass was submitted for pathological evaluation. The specimen contained an abscess characterized by necrotic tissue and a mixed inflam-

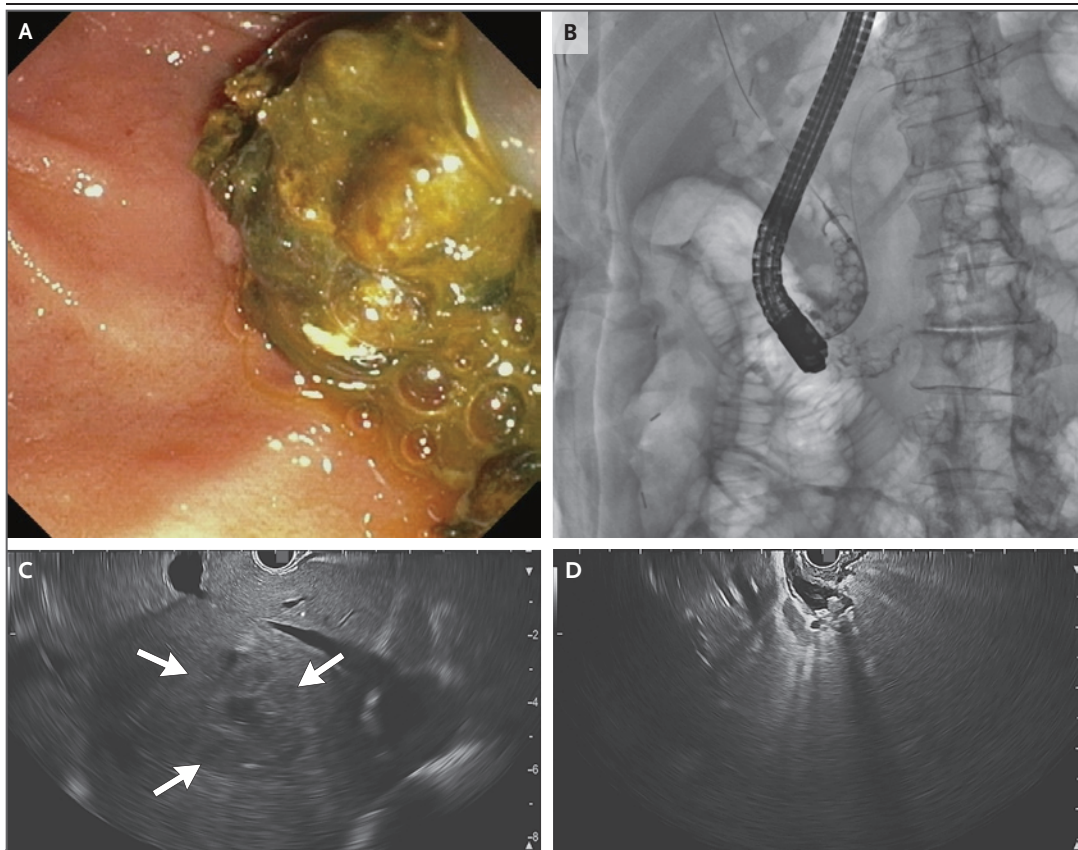


Figure 3. Endoscopic Images.

An image of the bile duct obtained during balloon-assisted upper endoscopy (Panel A) shows extensive sludge and small stone fragments. No fixed obstruction is seen in the main bile duct or in the intrahepatic branches. An initial endoscopic retrograde cholangiopancreatographic (ERCP) image (Panel B) shows multiple filling opacities consistent with gallstones, along with a multicystic liver mass. The final ERCP image (not shown) showed no obstruction, and hence, biliary stenting was not performed. Endoscopic ultrasound images (Panels C and D) show a heterogeneous, well-circumscribed mass in the left lobe of the liver (Panel C, arrows) and multiple gallstones in the bile duct (Panel D). Note that the gray box at the top center of Panels C and D obscures a small artifact.

matory infiltrate. Hematoxylin and eosin staining (Fig. 4A and 4B) revealed areas of necrosis with an infiltrate composed of neutrophils, lymphocytes, and macrophages that was consistent with acute inflammation. These findings strongly suggested an infection, although the specific causative microorganism remained undetermined.

Several conditions can have histologic features similar to these. Hepatic cancers such as hepatocellular carcinoma and metastatic tumors may also manifest with necrosis and inflammatory infiltrates but are typically characterized by more irregular margins, cellular atypia, and architectural distortion. In addition, granulomatous diseases such as

tuberculosis and sarcoidosis can manifest with granulomas in the liver; these were not observed in this case, which further supports the diagnosis of a nongranulomatous abscess. Ziehl–Neelsen staining (Fig. 4C) for acid-fast bacilli was negative, which argues against mycobacterial infection.

Fungal infections, such as those caused by candida or aspergillus species, can result in necrotic lesions with inflammatory infiltrates. However, periodic acid–Schiff staining with diastase and Grocott–Gomori methenamine silver staining (Fig. 4D and 4E) did not show definitive fungal elements.

Gram's staining, performed with the use of the

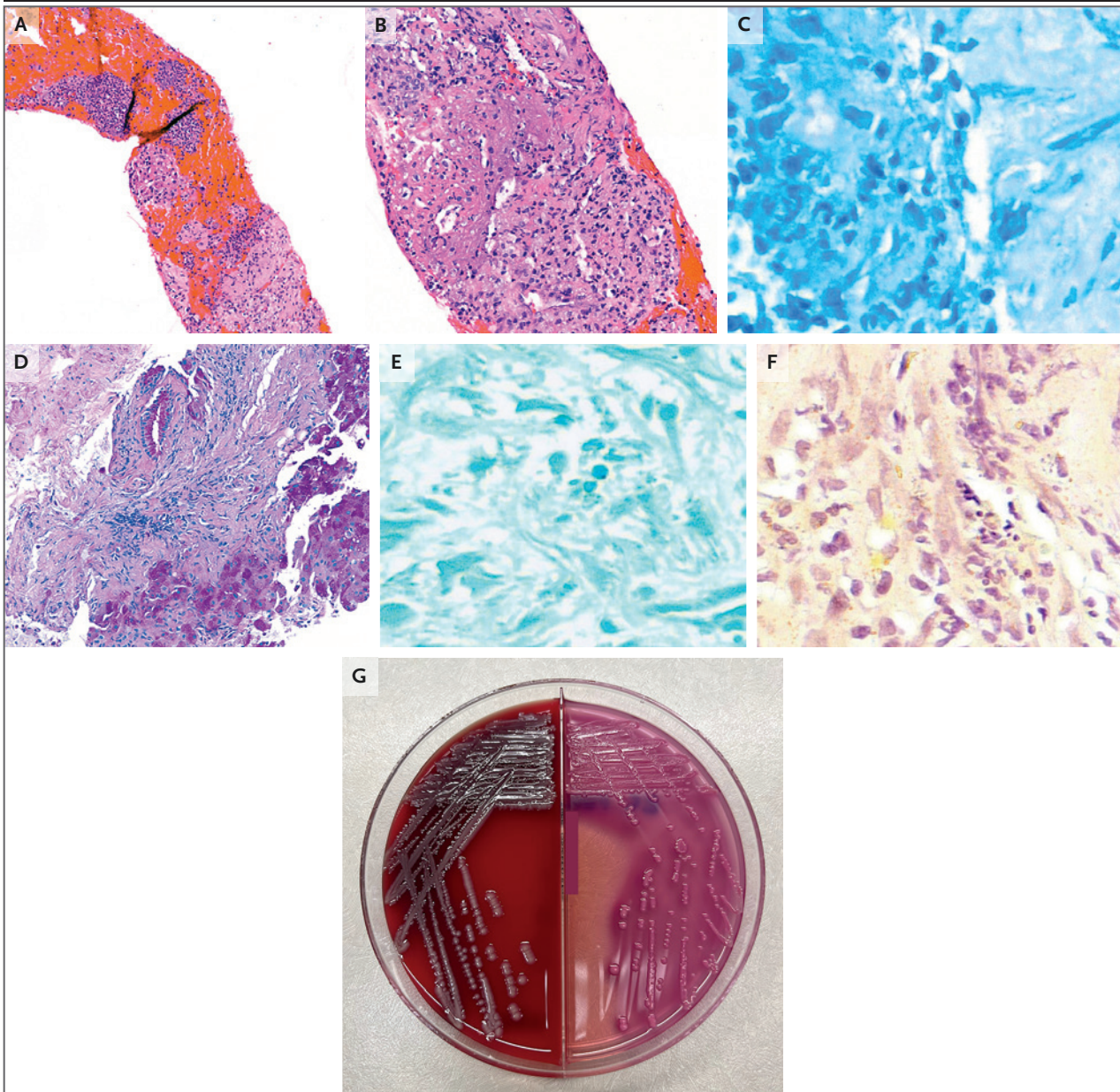


Figure 4. Biopsy Specimen of the Liver.

Hematoxylin and eosin staining (Panels A and B) shows necrotic liver parenchyma with a mixed inflammatory infiltrate, including neutrophils, lymphocytes, and macrophages (Panel A) and inflammatory changes consistent with an acute infectious process (Panel B). Ziehl–Neelsen staining of a biopsy specimen for acid-fast bacilli (Panel C) is negative. Periodic acid–Schiff staining with diastase (Panel D) shows no evidence of fungal organisms. Grocott–Gomori methenamine silver staining (Panel E) does not show fungal elements. Gram’s staining, performed with the use of the Brown–Hopps technique (Panel F), is negative for identifiable bacteria. On a biplate of blood agar and MacConkey agar (Panel G), bacterial growth is observed. The blood agar side of the plate supports the growth of gram-positive and gram-negative organisms but does not differentiate between them. On the MacConkey agar side, there is abundant growth of a gram-negative, lactose-fermenting organism (as evidenced by the pink color, which is produced by lactose fermentation on MacConkey agar), consistent with the growth of *Escherichia coli*.

Brown–Hopps method (Fig. 4F), can identify both gram-positive and gram-negative bacterial elements, but in this case, it did not identify a specific pathogen. Despite extensive pathological evaluation, no microorganisms were definitively detected, which highlights the need for further microbiologic evaluation to guide diagnosis and treatment.

ANATOMICAL DIAGNOSIS

Pyogenic liver abscess.

ADDITIONAL DISCUSSION OF MANAGEMENT

Dr. Kristen M. Hysell: The biopsy results were consistent with pyogenic liver abscess, and the infectious disease service was consulted for further management. We sought to identify possible sources of this infection and to consider the most likely pathogens.

Pyogenic liver abscesses commonly arise from the biliary tract owing to conditions such as cholangitis, biliary obstruction (e.g., stricture or gallstones), or direct extension from adjacent infected structures. Other sources include portal venous spread associated with gastrointestinal conditions (e.g., peritonitis, diverticulitis, appendicitis, and colon cancer), hematogenous spread from the systemic circulation (e.g., endocarditis and bacteremia), and trauma or procedures such as liver biopsy.

In this patient, imaging revealed choledocholithiasis with mild intrahepatic and extrahepatic ductal dilatation, which suggested a biliary tract origin. Enteric bacteria — which typically cause polymicrobial infection, with both aerobic and anaerobic organisms — were suspected. Common pathogens include enterobacterales species (e.g., *Escherichia coli* and *K. pneumoniae*), streptococcus species (especially the *Streptococcus anginosus* group, previously known as the *S. milleri* group), and enterococcus species. Anaerobes such as bacteroides and fusobacterium species are also frequently associated with pyogenic liver abscess but are often underreported owing to laboratory challenges with culture isolation.¹² *S. aureus* is a common cause of infection in cases involving hematogenous spread

or trauma; however, it would be unlikely to originate from the biliary tract. Fungal infections (e.g., infection with candida species) are less common in immunocompetent patients than in immunocompromised patients. In addition, we considered the possibility of infection with mycobacteria (e.g., *Mycobacterium tuberculosis*) or a parasite (e.g., echinococcus species and *E. histolytica*), but these infections would be rare in a patient without relevant risk factors.

Next, we aimed to prioritize pathogen identification. Blood specimens were obtained for culture because cultures can be positive in up to 50% of cases of pyogenic liver abscess; they were negative in this patient.^{12,13} Percutaneous CT-guided drainage of the abscesses was recommended for diagnostic and therapeutic purposes, enabling culture collection and improving source control. Adequate drainage can also potentially shorten the duration of antibiotic therapy.¹⁴ Because this patient's condition was clinically stable, antibiotics were not administered until aspiration was performed, with the hope of optimizing the culture yield.

After the drainage procedure, empirical antibiotic therapy was initiated to target the most likely enteric aerobic and anaerobic organisms. The patient had not recently received treatment with antibiotics, and she had no known history of multidrug-resistant infections. Therefore, the treatment regimen could include a third- or fourth-generation cephalosporin plus metronidazole or a β -lactam– β -lactamase inhibitor (e.g., piperacillin–tazobactam or ampicillin–sulbactam). Once a patient's culture results are available, antibiotic therapy can be adjusted to target the specific pathogen. However, empirical anaerobic coverage is usually continued even without isolation of an anaerobic organism, since isolation of these organisms in culture can be challenging.¹²

Antibiotic therapy is typically continued for 4 to 6 weeks, depending on the severity of the infection and the patient's response to treatment. Longer courses may be indicated if undrained collections are present. Transition from intravenous to oral antibiotics can also be considered in patients with a good clinical response to drainage.¹⁵ Repeat imaging is often helpful for assessing the resolution of abscess; however, radiologic findings may lag behind the clinical response,

and ultimately, the duration of treatment should be guided by the patient's clinical response.¹⁶

ADDITIONAL PATHOLOGICAL DISCUSSION

Dr. Alkashash: The aspirate obtained from the liver was cultured on both blood agar and MacConkey agar. Blood agar provides a rich medium that supports the growth of a broad range of microorganisms, including common gram-positive and gram-negative organisms. In contrast, MacConkey agar is selective for gram-negative bacteria, particularly enteric organisms. The use of both media enhances the likelihood of detecting the spectrum of possible pathogens.

Bacterial growth was observed on a biplate of blood agar and MacConkey agar (Fig. 4G), a finding that suggested the presence of a mixed microbial population. Matrix-assisted laser desorption ionization–time-of-flight (MALDI-TOF) mass spectrometry was subsequently used for precise microbial identification; this technique enabled the accurate detection of *E. coli* as the causative organism. The identification of *E. coli* in this case suggested either a gastrointestinal or hepatobiliary source of infection or hematogenous or portal venous spread from distant viscera, since this bacterium is a common pathogen in intraabdominal infections.

PATHOLOGICAL DIAGNOSIS

Pyogenic liver abscess due to *Escherichia coli* infection.

FOLLOW-UP

Dr. Hysell: After the diagnostic procedure, the patient initially received treatment with piperacillin–tazobactam, which was later changed to ceftriax-

one and metronidazole on the basis of results of antimicrobial susceptibility testing. After clinical improvement, her treatment regimen was transitioned to oral amoxicillin–clavulanate, and she was discharged home with instructions to continue antibiotic therapy and with plans for close outpatient follow-up.

One month after discharge, the patient presented with nausea and persistent poor oral intake, without fever, chills, abdominal pain, or vomiting. She reported intermittent nonadherence to her prescribed antibiotics, citing symptoms as the reason for missed doses. Repeat CT, which was performed to evaluate any changes since the previous imaging study, revealed residual inflammation and persistent edema but no new lesions or drainable fluid collections. Treatment with oral amoxicillin–clavulanate was switched to a 6-week course of oral ciprofloxacin and metronidazole, which caused fewer symptoms than the previous regimen. A follow-up CT scan showed further improvement, and by the end of the antibiotic course, the patient had excellent clinical resolution; her appetite was restored, her weight had returned to the baseline level, and symptoms had resolved.

FINAL DIAGNOSIS

Multifocal pyogenic liver abscess due to *Escherichia coli* infection.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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